Ladera Vista Invitational - Repaired practice edition - Answer key

Division B | Heredity | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (4 points) Authored written substitute for unavailable source material: Contrast an inducible and a repressible operon, with an example of each.

Unit: Gene expression: transcription and translation.
Worked-case data: A DNA template strand reads 3′-TAC AAA CCG ATT-5′. Give the mRNA and the amino acid sequence.
Separate investigation observations:
Medium: The bacteria grow with lactose and no glucose.
Repressor: The lac repressor is not bound to the operator.
cAMP: cAMP levels are high, and CAP is bound near the promoter.

answer: An inducible operon, like lac, is normally off and turns on when an inducer (allolactose) is present. A repressible operon, like trp, is normally on and turns off when a corepressor (tryptophan) builds up. (4 points).